Please check the examination details below before entering your candidate information

Candidate surname	Other names

Pearson Edexcel
Level 1/Level 2 GCSE (9–1)

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Half Past Paper – Week 8

Total Marks

Q	Topic		Sparx Code
2a	Solving direct proportion word problems	3	U721
2b	Solving direct proportion word problems	2	U721
4	Plans and elevations	2	U743
6	Sharing amounts in a given ratio	4	U577
8a	Estimating calculations	2	U225
8b	Multiplying and with place value and with standard form, Index rules with positive indices	1	U735,U264, U235
8c	Index rules with negative indices	1	U694
10	Solving simultaneous equations graphically	2	U836
12	Percentage calculations, Converting between fractions, decimals and percentages	3	U671,U278, U888
14	Using the exact values of trigonometric ratios	2	U627
16a	Using the product rule for counting	2	U369
16b	Using the product rule for counting	2	U369
18a	Adding and subtracting surds	2	U872
18b	Rationalising denominators containing a single term	3	U707
20	Constructing direct proportion equations, Constructing inverse proportion equations	4	U407,U138
22	Tree diagrams for independent events, Constructing and solving simultaneous equations	5	U558,U137
	Total	40	

Answer ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

2 Deon needs 50 g of sugar to make 15 biscuits.

She also needs

three times as much flour as sugar
two times as much butter as sugar

Deon is going to make 60 biscuits.

(*a*) Work out the amount of flour she needs.

..... g
(3)

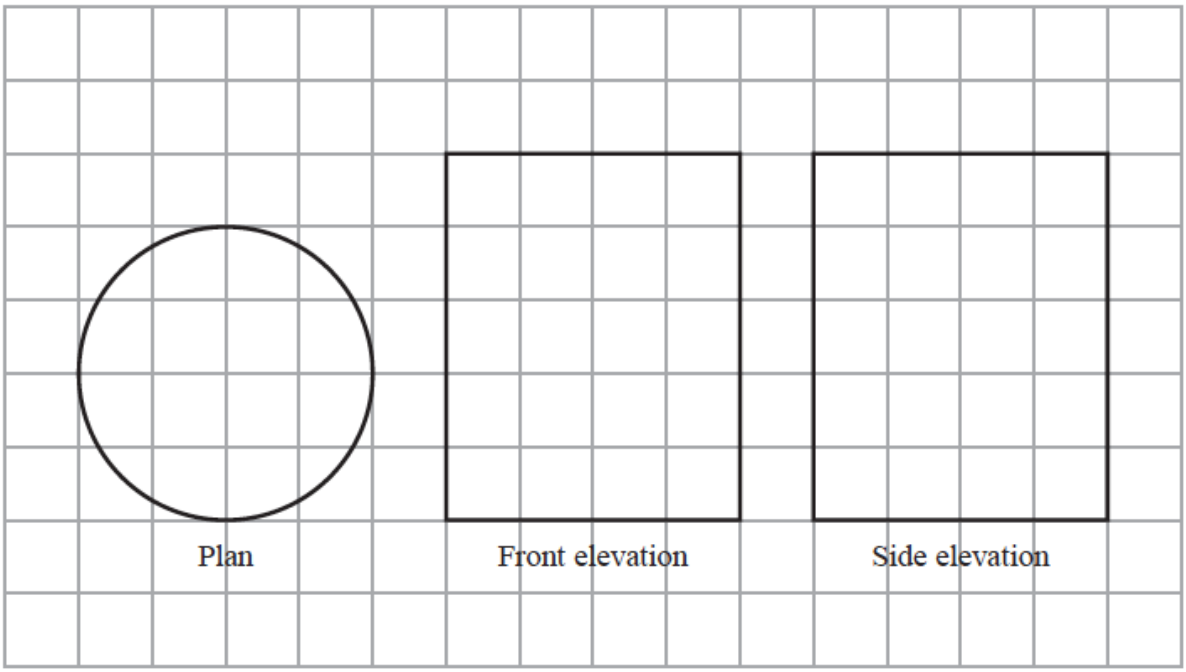
Deon has to buy all the butter she needs to make 60 biscuits.
She buys the butter in 250 g packs.

(*b*) How many packs of butter does Deon need to buy?

.....
(2)

(Total for Question 2 is 5 marks)

4 The diagram shows the plan, front elevation and side elevation of a solid shape, drawn on a centimetre grid.



In the space below, draw a sketch of the solid shape.
Give the dimensions of the solid on your sketch.

(Total for Question 4 is 2 marks)

6 A shop sells packs of black pens, packs of red pens and packs of green pens.

There are

2 pens in each pack of black pens
5 pens in each pack of red pens
6 pens in each pack of green pens

On Monday,

number of packs of black pens sold : number of packs of red pens sold : number of packs of green pens sold = 7 : 3 : 4

A total of 212 pens were sold.

Work out the number of green pens sold.

.....

(Total for Question 6 is 4 marks)

8 (*a*) Work out an estimate for the value of $\sqrt{63.5 \times 101.7}$

.....

(2)

$(2.3)^6 = 148$ correct to 3 significant figures.

(*b*) Find the value of $(0.23)^6$ correct to 3 significant figures.

.....

(1)

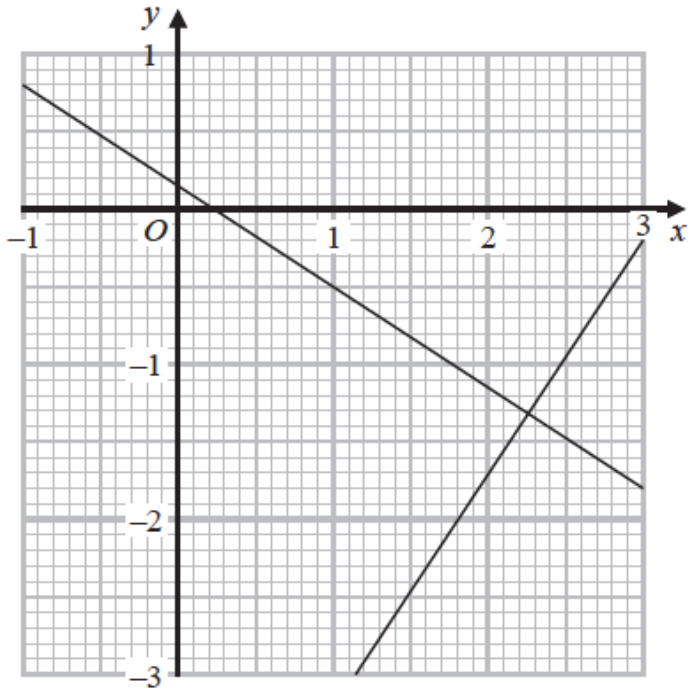
(*c*) Find the value of 5^{-2}

.....

(1)

(Total for Question 8 is 4 marks)

10 The graphs with equations $3y + 2x = \frac{1}{2}$ and $2y - 3x = -\frac{113}{12}$ have been drawn on the grid below.



Using the graphs, find estimates of the solutions of the simultaneous equations

$$3y + 2x = \frac{1}{2}$$

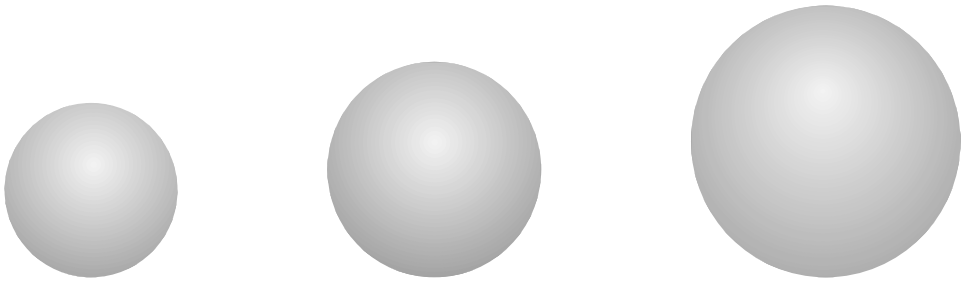
$$2y - 3x = -\frac{113}{12}$$

$x =$ .....

$y =$ .....

(Total for Question 10 is 2 marks)

12 Here are three spheres.



P **Q** **R**

The volume of sphere **Q** is 50% more than the volume of sphere **P**.
The volume of sphere **R** is 50% more than the volume of sphere **Q**.

Find the volume of sphere **P** as a fraction of the volume of sphere **R**.

.....

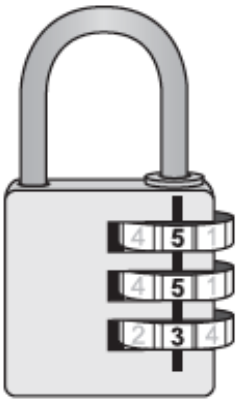
(Total for Question 12 is 3 marks)

14 Find the exact value of $\tan 30° \times \sin 60°$
Give your answer in its simplest form.

.....

(Total for Question 14 is 2 marks)

16 There are three dials on a combination lock.
Each dial can be set to one of the numbers 1, 2, 3, 4, 5
The three digit number 553 is one way the dials can be set, as shown in the diagram.



(*a*) Work out the number of different three digit numbers that can be set for the combination lock.

.....

(2)

(*b*) How many of the possible three digit numbers have three different digits?

.....

(2)

(Total for Question 16 is 4 marks)

18 (*a*) Express $\sqrt{3}+\sqrt{12}$ in the form $a\sqrt{3}$ where a is an integer.

.....

(2)

(*b*) Express $\left(\frac{1}{\sqrt{3}}\right)^7$ in the form $\frac{\sqrt{b}}{c}$ where b and c are integers.

.....

(3)

(Total for Question 18 is 5 marks)

20 h is inversely proportional to p

p is directly proportional to $\sqrt{t}$

Given that $h = 10$ and $t = 144$ when $p = 6$
find a formula for h in terms of t

.....

(Total for Question 20 is 4 marks)

22 There are only r red counters and g green counters in a bag.

A counter is taken at random from the bag.

The probability that the counter is green is $\frac{3}{7}$

The counter is put back in the bag.

2 more red counters and 3 more green counters are put in the bag.
A counter is taken at random from the bag.

The probability that the counter is green is $\frac{6}{13}$

Find the number of red counters and the number of green counters that were in the bag originally.

red counters.....

green counters.....

(Total for Question 22 is 5 marks)

TOTAL FOR PAPER IS 40 MARKS